

Multivariable Calculus Final Project

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Documentation

Introduction & Design

For this project, our group chose a cone with radius 30" and height 36" (or 2.5' and 3', respectively). The equation that we used was

$$\frac{x^2}{900} + \frac{y^2}{900} = \frac{z^2}{1296},$$

where $x \in [-30, 30]$, $y \in [-30, 30]$, and $z \in [0, 36]$.

Calculations

Theoretical Precise Value

To calculate the precise theoretic value, we will use the triple integral

$$\iiint_Q f(x, y, z) dV.$$

To find the bounds, we will use cylindrical coordinates, as Cartesian coordinates tend to end up messy in such calculations. To find the bounds in cylindrical coordinates, we only need to solve for r , as $\theta \in [0, 2\pi]$ and $z \in [0, 36]$, as given by our domain restrictions above.

To find the bounds for r , we first convert the rectangular x and y to cylindrical coordinates. Following the conversion, our cone can be represented by the equation

$$\frac{r^2}{900} = \frac{z^2}{1296}.$$

Solve for r :

$$r = \sqrt{\frac{900z^2}{1296}} = \sqrt{\frac{25z^2}{36}} = \frac{5z}{6}.$$

Thus, the upper bound for r is $\frac{5z}{6}$, and because the lower bound for r is 0, we know that the complete bounds for r are $[0, \frac{5z}{6}]$. Thus, the complete bounds, for all variables, are:

$$\theta \in [0, 2\pi], \quad z \in [0, 36], \quad \text{and} \quad r \in \left[0, \frac{5z}{6}\right].$$

Therefore, the volume integral for the cone is If we plug in all the values for the triple integral, we end up with

$$\int_0^{2\pi} \int_0^{36} \int_0^{\frac{5z}{6}} r \, dr \, dz \, d\theta,$$

which we can solve as follows:

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^{36} \int_0^{\frac{5z}{6}} r \, dr \, dz \, d\theta = \int_0^{2\pi} \int_0^{36} \left[\frac{r^2}{2} \right]_0^{\frac{5z}{6}} dz \, d\theta = \int_0^{2\pi} \int_0^{36} \frac{25z^2}{72} dz \, d\theta \\ &= \int_0^{2\pi} \left[\frac{25z^3}{216} \right]_0^{36} d\theta = 5040 \int_0^{2\pi} d\theta = 5040 (2\pi) = 10800 (2\pi) = 33929.2007. \end{aligned}$$

Thus, the theoretical precise volume of our cone is 33929.2007 cubic inches (or approximately 19.635 cubic feet).